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Association between genetically predicted circulating immune cells on periodontitis highlights the prospect of systemic immunomodulation management in periodontal care: a Mendelian randomization study

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Abstract

Background

Periodontitis drives irreparable destruction of periodontal tissue and possesses a tendency to aggravate inflammatory disorders. Systemic immunomodulation management remains an attractive approach in periodontal care within a context of “predictive, preventive, and personalized” periodontics. Leveraging a comprehensive Mendelian randomization (MR), the present study aims to evaluate the causal relationships between circulating immune cells and the risk of periodontitis.

Methods

Genetic proxies for circulating immune cells and periodontitis were obtained from genome-wide association studies. We performed a two-sample bidirectional univariable Mendelian Randomization, followed by sensitivity testing, multivariable MR, subgroup analysis, and Bayesian model averaging (MR-BMA) to explore the causal association between them. The transcriptome-wide association study (TWAS) and colocalization analysis were further conducted to identify responsible genes as immunomodulation candidates for periodontal care.

Results

MR study indicated that circulating neutrophils, Natural Killer T cells, and plasmacytoid Dendritic Cells were relative to a higher risk of periodontitis, with no evidence of heterogeneity or pleiotropy. The MR-BMA identified neutrophils as the primary factor responsible for periodontitis. The TWAS recognized five cross-trait genes to be involved in their interaction. Two high-confidence genes on 1q21.3, *S100A9* and *S100A12*, could represent immunomodulation targets for neutrophils in periodontitis.

Conclusions

The present study suggests the causative association between circulating immune cells and the risk of periodontitis. Our findings highlight the prospect of systemic immunomodulation management in periodontal care, which can be valuable for early diagnostics, risk assessment, targeted prevention, and personalized management of periodontitis. More research is required to comprehend the biological plausibility and clinical applicability of our findings.

Funding

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The authors analyzed the causative association between circulating immune cells and periodontitis, and reported three risk immune cells related to periodontitis. The significance of the findings is **fundamental**, which substantially advances our understanding of periodontitis. The strength of evidence is **convincing**.

Introduction

Periodontitis imposes a considerable social burden on dental practice and general health

Periodontitis is a highly prevalent disease that affects a considerable percentage of the population. According to large-scale epidemiological research, up to half of all adults worldwide suffer periodontal disease, with severe periodontitis threatening 10.5-12.0 % of them ([Kassebaum et al., 2014](#)). Furthermore, periodontitis is the leading cause of adult tooth loss, necessitating extensive dental procedures such as extractions, dental implants, or prosthetics, which can be costly and time-consuming for both patients and dental practitioners ([Genco and Sanz, 2020](#)). Recent research demonstrated a relationship between periodontitis and inflammatory comorbidities such as type 2 diabetes, cardiovascular disease, rheumatoid arthritis, and inflammatory bowel disease

([Hajishengallis and Chavakis, 2021](#)). The high prevalence and harmful implications of periodontitis underline the importance of managing periodontitis to maintain oral and general health ([Peres et al., 2019](#)). Since early-stage prevention is the most significant way to improve health, the identification of additional potential risk factors was required to provide predictive, preventive, and personalized strategies for periodontal care ([Ma et al., 2021](#)).

Evidence from epidemiology and pathophysiology demonstrates the impact of circulating immune cells on periodontitis

Periodontitis is a chronic infectious disease characterized by the interactions between microorganisms and host immune response ([Curtis et al., 2020](#)). The immune response to periodontitis comprises both innate and adaptive immunity, with multiple cytokines, immune cells, and inflammatory pathways participating in a complex interplay ([Dutzan et al., 2016](#)). Systemic immunological alternations, such as circulating immune cells, play a crucial role in the initiation and progression of periodontitis ([Cekici et al., 2014](#)). An observational study indicated that patients with periodontitis experience a greater level of circulating leukocytes ([Noz et al., 2021](#)), while another discovered that the distribution of B cells alters in the context of severe periodontitis, with a higher proportion of circulating memory B cells ([Demoersman et al., 2018](#)). Furthermore, inflamed periodontal tissue recruits immune cells from circulation ([Hajishengallis, 2020](#)). As reported, the number of immune cells in periodontal tissue changes as periodontitis progresses, featuring an increase in monocytes, and B cells and a decrease in T cells ([Nair et al., 2014](#); [Steinmetz et al., 2016](#)). The promising concept of “trained immunity” has recently provided a greater understanding of the host immune response in periodontitis ([Netea et al., 2020](#)), which can explain the fact that the increased hyper-responsiveness of circulating immune cells from patients with periodontitis as well as its probable mechanism of mediating periodontitis and its comorbidities ([Li et al., 2023](#)).

Immunomodulation of systemic immune response serves as a hub for periodontal care

Systemic immunomodulation management has the potential to improve host homeostasis by altering the composition and function of the immune milieu ([Yang et al., 2021](#)). Periodontitis can be effectively managed by restricting immune cell activation, implying that immunomodulators have significant promise in constructing comprehensive strategies for periodontal management ([Zidar et al., 2021](#)). For example, resveratrol, quercetin, and N-acetylcysteine were reported to reduce the release of reactive oxygen species (ROS) by neutrophils, which aided in the prevention of periodontitis ([Orihuela-Campos et al., 2015](#)). Nonetheless, from a medical and therapeutic perspective, it is critical to determine whether the link between circulating immune cells and periodontitis is merely correlative or driven by causative mechanistic interactions ([Lamont and Hajishengallis, 2015](#)). Understanding the role of systemic immune alternations in periodontitis is critical for developing an effective strategy for early screening of high-risk patients, prompt implementation of definitive prevention, and individualized deployment of targeted treatment, all with the goal of reducing unexpected inflammatory responses, maintaining oral health, and avoiding complications ([Zhang et al., 2023](#)).

Mendelian randomization is a powerful complement to causal inference in terms of genetics

Previous research has substantiated the potential of immunomodulation management in predicting and preventing periodontitis; however, in observational studies, the association is frequently disguised by reverse causality, confounding factors, and disease conditions, which obscured the intrinsic causal inference between them ([Hajishengallis and Korostoff, 2017](#)). Mendelian randomization (MR) investigates the causal relationships between risk factors and

diseases by exploiting genetic variants as instrumental variables (IVs) (Davies et al., 2018). which is less likely to be affected by underlying bias or disease condition, in that alleles are randomly allocated from parents to offspring (Julian et al., 2023). Notably, MR with distinct causal relationships may provide fresh evidence from a genomics perspective (Golubnitschaja et al., 2014). We postulate that individuals with a disproportionate immunological network have a higher risk of periodontitis due to unexpected inflammatory reactions.

Methods

Study design

The present study, as shown in **Figure 1**, was based on the Strengthening the Reporting of Observational Studies in Epidemiology using the Mendelian Randomization (STROBE-MR) checklist (Skrivankova et al., 2021). The present research aims to evaluate the causal association between circulating immune cells and the risk of periodontitis, providing insight into opportunity for systemic immunomodulation management in periodontal care.

Overall, we used summary statistics from publicly accessible genome-wide association studies (GWASs) to conduct univariable Mendelian randomization (UVMR) and multivariable Mendelian randomization (MVMR) analyses. We replicated UVMR analysis after removing potential outlier, influential, or pleiotropic SNPs, as well as a performed subgroup and reverse MR. Then, the Bayesian model averaging (MR-BMA) was employed to pinpoint the predominant characteristics with causal signals. Finally, we conducted a transcriptome-wide association study (TWAS) and colocalization analysis to identify potential genes implicated in biological connections.

Data source

Summary-level data on 17 circulating immune cells were obtained from the large-scale GWAS conducted by the Blood Cell Consortium (BCC) and the Sardinian cohort (Orrù et al., 2020; Vuckovic et al., 2020). The GWAS data for periodontitis and its subtypes were supplied by the Gene-Lifestyle Interactions in Dental Endpoints collaboration consortium (GLIDE) and the FinnGen cohort (Kurki et al., 2023; Shungin et al., 2019). Of note, we did a screening of the population to ensure homogeneity within the target group and to reduce overlapping between them. For example, the population from the Latin American and the UK Biobank cohort were omitted from the periodontitis data set. Of note, we filtered the population to ensure homogeneity and to eliminate overlaps within the target group. The population from the Latin American and UK Biobank cohorts, for example, were excluded from the data set of periodontitis. The features of GWAS characteristics and included cohorts were highlighted in **Table 1** and **Supplementary File 1** — Table s1.

Candidates for IVs underwent a thorough set of screening procedures. To equalize the sample disparities among databases, a complicated screening criterion was performed. We initially filtered the *p*-values of the single nucleotide polymorphisms (SNPs), followed by the selection of independent SNPs using the linkage disequilibrium (LD) approach. The R^2 and F-statistics were introduced to demonstrate the degree of genetic variation explained and their relative impact on the outcomes, and SNPs with F-statistics < 10 would be removed based on the first MR assumption (Papadimitriou et al., 2020). In addition, SNPs that exhibited a direct association with the outcome would also be deleted to support the third MR assumption ($P < 5 \times 10^{-8}$). Throughout the harmonization processes, palindromic and ambiguous SNPs were eliminated to ensure the reliability and validity of causal inference. In MVMR, we excluded SNPs situated in the major histocompatibility complex area (MHC, 6p21.31) due to its complexity and confounding effects (Burgess and Thompson, 2015).

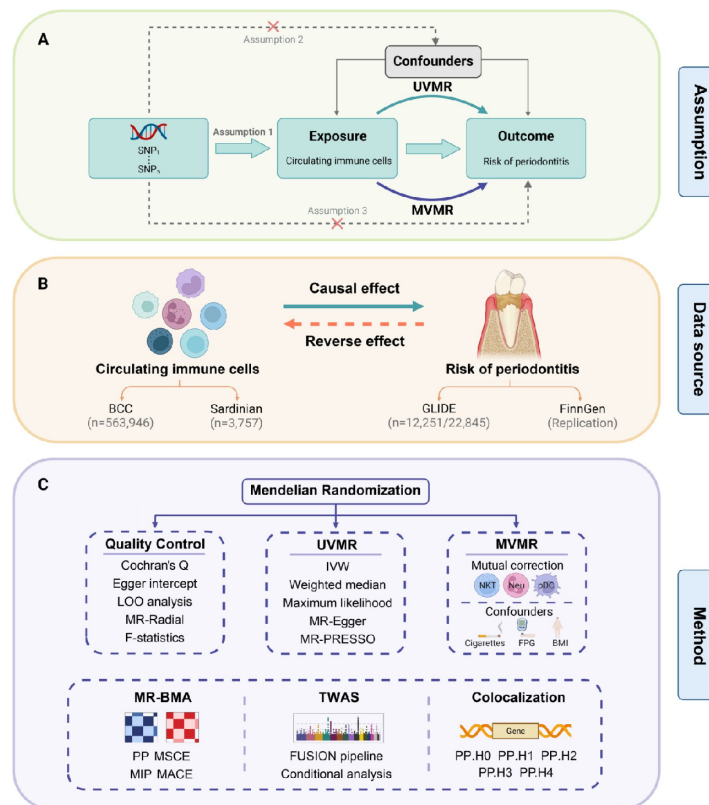


Figure 1.

Study design. **(A)** Overview of the process and principal assumptions of MR. **(B)** Data sources of the GWASs. **(C)** Methods performed in the present study.

Abbreviations and Notes: BCC, Blood Cell Consortium; BMA, Bayesian model averaging, a high-throughput method based on non-linear regression; BMI, body mass index; FPG, fasting plasma glucose; FUSION, functional summary-based imputation; GLIDE, Gene-Lifestyle Interactions in Dental Endpoints collaboration consortium; GWAS, genome-wide association study; IVW, inverse variance weighted, the primary method in MR to explore the association between exposure and outcome; LOO, leave-one-out, a method for detecting potential influential SNPs; SNP, single nucleotide polymorphism, as genetic instrumental variables for the exposure and outcome; MACE, model-averaged causal estimate; MIP, marginal probability of inclusion; MR, Mendelian randomization; MR-PRESSO, Mendelian Randomization Pleiotropy RESidual Sum and Outlier, a method for assessing and rectifying pleiotropic SNPs; MSCE, model-specific causal estimate; MVMR, multivariable Mendelian randomization, a MR model for adjusting confounding and mutual correction; Neu, neutrophil; NKT, Natural Killer T cell; pDC, plasmacytoid dendritic cell; PP, posterior probability; TWAS, transcriptome-wide association study; UVMR, univariable Mendelian randomization.

Phenotype	Year	Sample size (ncase/ncontrol)	nSNP	Ancestry	Unit	Consortium/cohort	PMID
<i>Exposure</i>							
Circulating immune cells	2020	563,946	15million	European	/nL	BCC	32888494
Lymphocyte subsets	2020	3,757	15.2million	European	/ug	Sardinian cohort	32929287
<i>Outcome</i>							
Periodontitis	2019	35,096 (12,251/22,845)	10.8million	European	Event	GLIDE	31235808
Chronic periodontitis (FinnGen)	2023	263,668 (4,434/259,234)	20.2million	European	Event	FinnGen (R9K11)	36653562
Gingival hyperplasia (FinnGen)	2023	259,613 (379/259,234)	20.2million	European	Event	FinnGen (R9K11)	36653562
<i>Covariate</i>							
Cigarettes smoked per day	2019	249,752	12million	European	1/SD	GSCAN	30643251
Fasting plasma glucose	2021	200,622	31million	European	mmol/L	MAGIC	34059833
Body mass index	2018	681,275	2.3million	European	kg/m ²	GIANT	30124842

Abbreviations: BCC, Blood Cell Consortium; GIANT, GWAS of the Genetic Investigation of ANthropometric Traits; GLIDE, Gene-Lifestyle Interactions in Dental Endpoints; GSCAN, GWAS and Sequencing Consortium of Alcohol and Nicotine use; MAGIC, Meta-Analysis of Glucose and Insulin-related Traits Consortium; SD, standard deviation; SNP, single nucleotide polymorphism.

Table 1.

Characteristics of the GWAS data for MR.

Univariable and multivariable Mendelian randomization

In UVMR, the inverse-variance weighted (IVW) method was performed as the primary analysis, and four alternative MR methods, including weighted median, maximum likelihood, MR-Egger, and MR pleiotropy residual sum and outlier (MR-PRESSO) global test were employed for sensitivity testing to assess the robustness of the IVW estimates. Briefly, the IVW assumes that all genetic variations meet the conditions and integrated estimates from multiple genetic variants by weighting them inversely to variances ([Sanderson et al., 2022](#)). The weighted median generates precise estimates when more than half of the SNPs are valid ([Gormley et al., 2023](#)). The maximum likelihood offers a normal bivariate distribution to estimate causal effects by maximizing the likelihood function with a linear relationship ([Xue et al., 2021](#)). MR-Egger provides estimates after accounting for possible horizontal pleiotropy discovered by its incorporated intercept test, albeit the estimates were frequently underpowered ([Bowden et al., 2016](#)). MR-PRESSO detects outliers that cause pleiotropy and generates estimates once these outlier SNPs are eliminated ([Verbanck et al., 2018](#)).

To gauge the individual influence of each variant, MVMR analysis with mutual adjustment was performed, followed by a correction for associated confounders or intermediates ([Burgess and Thompson, 2015](#)). As the main test, the MVMR-IVW method, offered by the MVMR-least absolute shrinkage and selection operator (MVMR-LASSO), and the MVMR-Egger method were chosen ([Bowden et al., 2016](#)). Notably, the MVMR-LASSO regression produces reliable estimations for moderate-to-high degrees of heterogeneity or pleiotropy, as well as aids in mitigating the potential effects of multicollinearity among the variables ([Grant and Burgess, 2021](#)).

The heterogeneity and horizontal pleiotropy of the results were quantified using Cochran's Q statistics and the intercept term in MR-Egger regression, respectively. The MR-Radial, a more sensitive method for outliers, would be employed to detect and remove outlier SNPs whenever heterogeneity or pleiotropy was discovered ([Bowden et al., 2018](#)). The leave-one-out (LOO) analysis and scatter plot were carried out to detect influential SNPs. A bilateral $P < 0.05$ was used as the threshold for statistical significance.

Bayesian model averaging

As a multivariate framework for high-throughput risk factors based on non-linear regression, the MR-BMA was then employed to explore the leading traits responsible for outcome ([Zuber et al., 2020](#)). First, we used closed-form Bayes factors and independence priors to calculate the posterior probability (PP) and model-specific causal estimates (MSCE) of each variant. Next, the total PPs for all potential models were added up to determine the marginal probability of inclusion (MIP). The model-averaged causal estimate (MACE), which reflected the average direct effect of each metabolic trait on the outcomes, was also used to compare risk factors and interpret the directions. Finally, the best model was chosen preferably based on the ranking of each model's MIP and PP values. The Q-statistic and Cook's distance were used to identify invalid outliers and influential variants within the model. The MR-BMA would be repeated once unqualified variations were discovered ([Eledum, 2021](#)).

Transcriptome-wide association study

We exploited the updated Genotype-Tissue Expression (GTEx) project Version 8 whole-blood data for transcriptome-wide association study (TWAS) analysis ([Gusev et al., 2016](#)). First, the functional summary-based imputation (FUSION) pipeline was used to infer the transcriptome associated with significant outcomes, among which the optimal gene expression model was chosen by comparing the values of R^2 provided by Bayesian sparse linear mixed models and multiple penalized linear regressions. A Bonferroni-corrected criterion of $P < 6.27 \times 10^{-6}$ (0.05/7,890 genes) was adopted as a measure of statistical significance. Then, conditional analysis and permutation testing were implemented to assess the dependability and robustness of the gene transcript-trait relationships discovered through TWAS. Finally, we performed a expression

quantitative trait locus (eQTL) colocalization analysis on TWAS-derived genes to determine whether the association was caused by a single causal SNP (PP.H4) or distinct causal SNPs (PP.H3). $PP.H3 + PP.H4 > 0.8$ was considered significant evidence of colocalization (Wallace, 2020 [DOI](#)).

Statistical analyses

Two-sided t-tests were used for statistical analyses. Odds ratios (OR) with 95% confidence intervals (CI) were utilized to measure the influence of circulating immune cells on the likelihood of periodontitis. All statistical analyses were performed using “TwoSampleMR” (version 0.5.7), “MRPRESSO” (version 1.0), “MendelianRandomization” (version 0.7.0), “RadialMR” (version 1.1), “coloc” (version 5.1.0.1), and “MVMR” (version 0.4) packages in R software (version 4.3.1).

Results

Selection of instrumental variables

A rigorous threshold of $P < 1 \times 10^{-9}$ was applied to the database with an abundance of positive SNPs (as in the BCC consortium) to ensure the reliability of IVs. Otherwise, a relatively strict standard of $P < 1 \times 10^{-6}$ was initially adopted (as in the Sardinian cohort), and we would loosen it at $P < 5 \times 10^{-6}$ if less than three SNPs met this threshold (a basic requirement for MR-PRESSO analysis). As a result, a total of 1940 SNPs were selected as IVs in the present study (**Supplementary File 1** — Table s2). The F-statistics ranged from 28.67 to 220.07, indicating a low risk of weak instrument bias.

Univariable Mendelian randomization

Three circulating immune cells were identified to be suggestively significant in the IVW method [OR: 1.09, 95% CI: 1.01-1.17, $P = 0.030$ for Natural Killer T cells (NKT); OR: 1.11, 95% CI: 1.00-1.23, $P = 0.042$ for neutrophils; OR: 1.13, 95% CI: 1.02-1.25, $P = 0.025$ for plasmacytoid Dendritic Cells (pDC)], which were further supported by the maximum likelihood and MR-PRESSO (**Figure 2A,2B** [DOI](#), **Supplementary File 1** — Table s3). The MR-Egger regression revealed no evidence of horizontal pleiotropy (p -values for intercept > 0.05). However, significant heterogeneity was detected in two traits (memory B cell and monocyte) (**Supplementary File 1** — Table s4), which faded after the removal of outliers (**Supplementary File 1** — Table s5, **Supplementary File 2** — Figure s1). Moreover, the LOO analysis showed no influential SNPs that were significantly linked with the outcome (**Supplementary File 2** — Figure s2). The observed significant results remained robust after removing pleiotropic SNPs (**Supplementary File 1** — Table s6, **Supplementary File 2** — Figure s3), and the scatter plot displayed a balance distribution amongst SNPs (**Figure 2C-2E** [DOI](#)).

Multivariable Mendelian randomization

After accounting for variable mutual adjustment and covariate correction for potential confounders [cigarettes smoked, fasting plasma glucose (FPG), and body mass index (BMI)], the causal relationship between circulating neutrophils and periodontitis remained stable with no evidence of heterogeneity or pleiotropy. NKT remained stable after modifying pDC and BMI, owing to the considerable heterogeneity of its impact on BMI; nonetheless, while the MR-LASSO analysis was used to make appropriate corrections, it was still advised to exercise caution when interacting with the effect. What’s more, pDC maintained stability after adjusting for NKT, whereas the strength of the observed association was compromised in the MR-Egger sensitivity analysis (**Figure 3** [DOI](#), **Supplementary File 1** — Table s7).

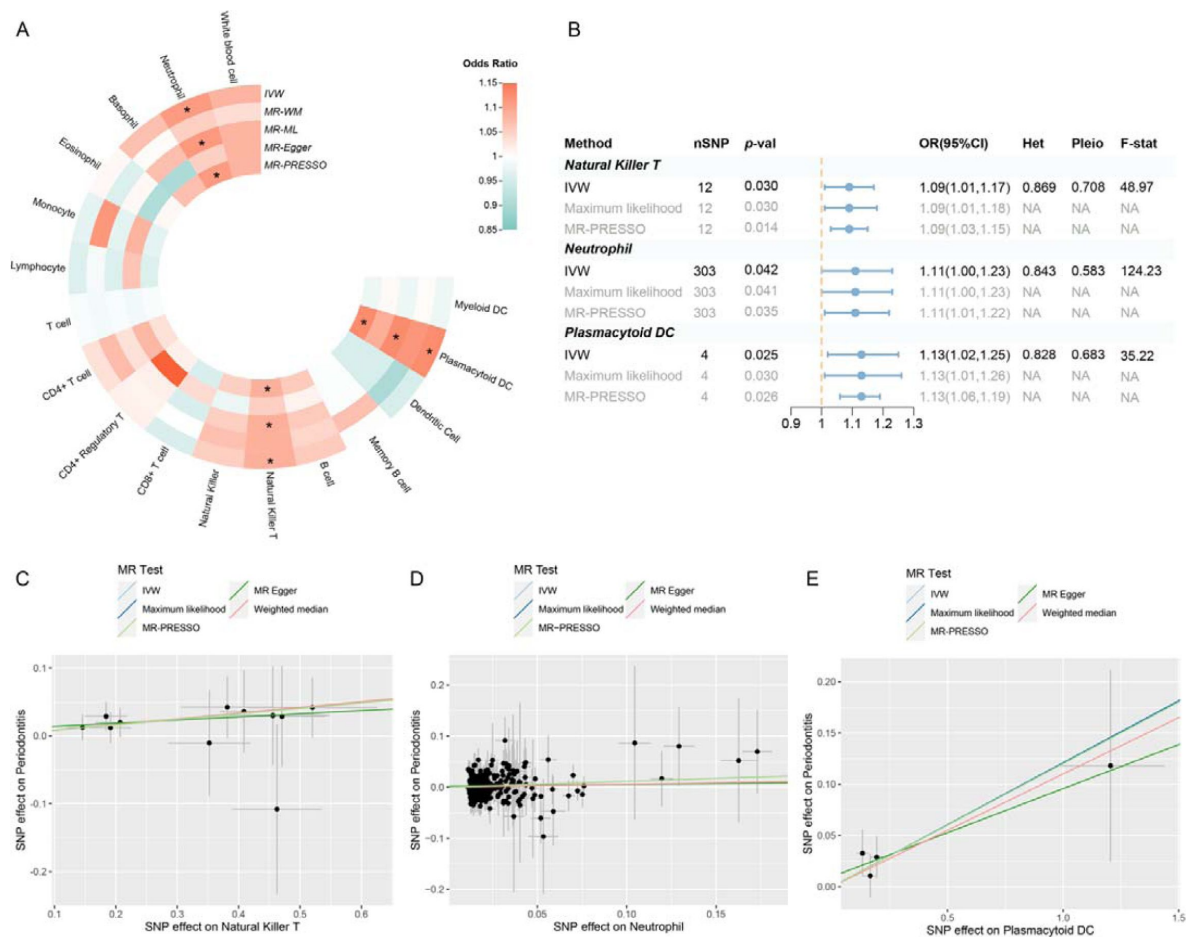


Figure 2.

Results of the UVMR. **(A)** A circular heatmap representing the MR analyses for the associations between circulating immune cells and the risk of periodontitis. Lines, from outermost to innermost, represent IVW, MR-WM, MR-ML, MR-Egger, and MR-PRESSO respectively. The color scale of the heatmap is based on the OR. * $P < 0.05$. **(B)** A forest plot of the MR analyses for significant results in [Figure 2A](#) ($P < 0.05$). The effects are quantified using OR with 95% CI. **(C-E)** The effect estimate for each individual variant in Natural Killer T cell **(C)**, neutrophil **(D)**, and plasmacytoid DC **(E)** is provided by plotting SNP-outcome associations against SNP-exposure associations. The regression slope fitted by different MR methods is represented by lines with different colors.

Abbreviations: CI, confidence interval; DC, dendritic cell; F-stat, F-statistic; IVW, inverse variance weighted; Het, heterogeneity; MR, Mendelian randomization; MR-ML, Mendelian randomization weighted median; MR-WM, Mendelian randomization maximum likelihood; MR-PRESSO, Mendelian Randomization Pleiotropy RESidual Sum and Outlier; OR, odds ratio; Pleio, pleiotropy; SNP, single nucleotide polymorphism; UVMR, univariable Mendelian randomization.

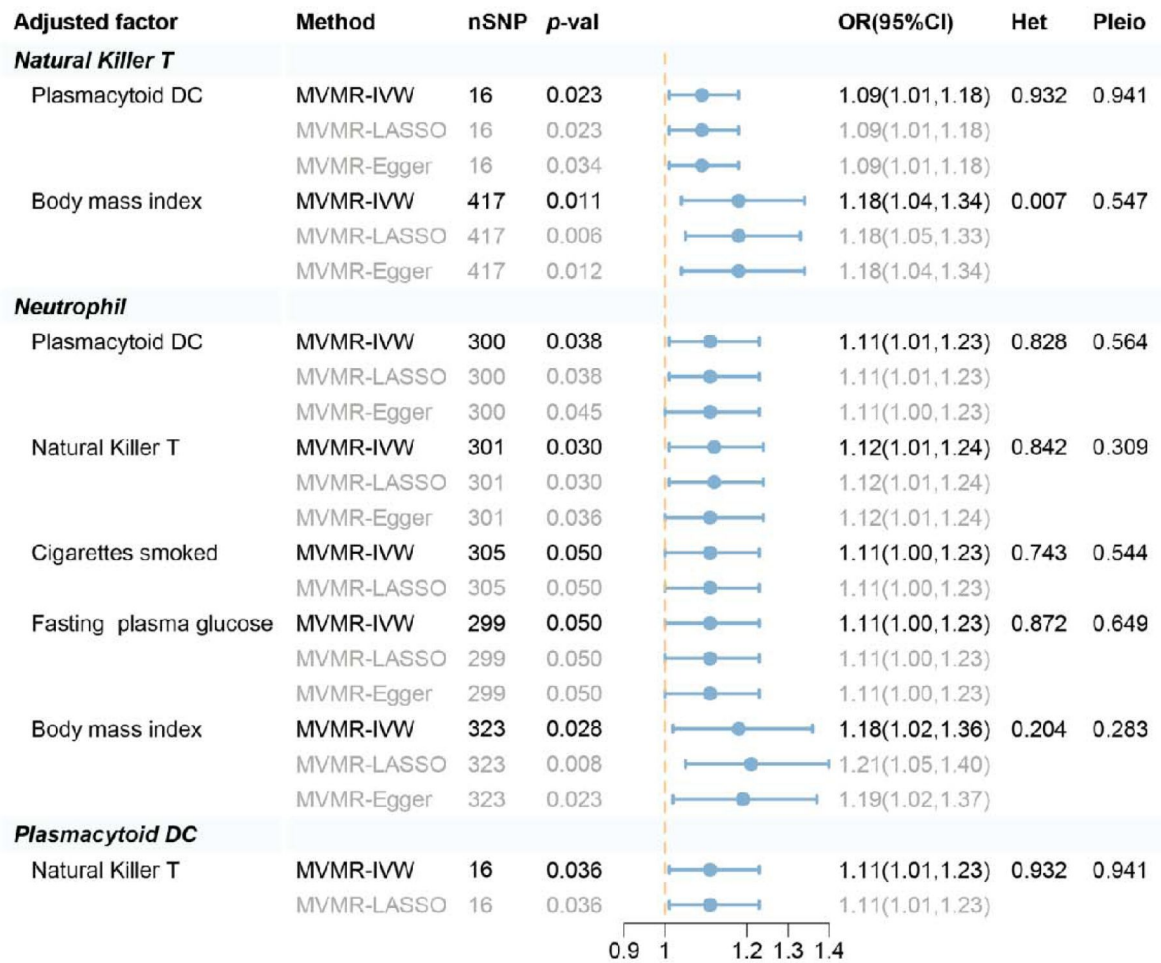


Figure 3.

Forest plot analysis of the MVMR of significant results following mutual adjustment and confounder correction. The effects are quantified using OR with 95% CI.

Abbreviations: CI, confidence interval; DC, Dendritic Cell; IVW, inverse variance weighted; Het, heterogeneity; LASSO, least absolute shrinkage and selection operator; MVMR, multivariable Mendelian randomization; OR, odds ratio; Pleio, pleiotropy; SNP, single nucleotide polymorphism.

Bayesian model averaging

The best risk models and factors were ordered and prioritized based on their PP and MIP (**Table 2**, **Supplementary File 1** — Table s8). Consequently, we observed that neutrophil was the best model and leading factor for periodontitis (PP=0.771, MSCE=0.108, MIP=0.895, MACE=0.097), followed by NKT and pDC. The Cochran's Q test and Cook's distance failed to detect outlier or influential variations (**Supplementary File 2** — Figure s4).

Subgroup analysis and reverse Mendelian randomization

A replication UVMR of two subgroups of periodontal diseases (chronic periodontitis and gingival hyperplasia) was also performed in the FinnGen cohort (**Figure 4**, **Supplementary File 1** — Table s9,10). The significant results revealed in the GLIDE database, however, could not be duplicated in the subgroup analysis, which may be attributed to the complexity of periodontal disease pathophysiology. Intriguingly, B cell was discovered to be involved in both subgroups of the FinnGen population (OR: 1.11, 95% CI: 1.02-1.22, $P = 0.019$ for chronic periodontitis; OR: 1.39, 95% CI: 1.02-1.88, $P = 0.036$ for gingival hyperplasia). Reverse MR revealed no indication of reverse causality (**Figure 4**, **Supplementary File 1** — Table s11).

Transcriptome-wide association study

The TWAS indicated that five cross-trait genes, including *CC2D2B* (10q24.1), *RP11-326C3.7* (11p15.5), *USP3* (15q22.31), *HERC1* (15q22.31), and *AMFR* (16q13), may be implicated in the interaction of circulating immune cells with periodontitis (**Figure 5A**, B). After Bonferroni correction ($P < 6.27 \times 10^{-6}$), we identified 658 of 3081 characteristics that were significantly associated with neutrophils, 5 of 443 with NKT, and 5 of 1038 with pDC. Within a broad criteria ($P < 5 \times 10^{-4}$), we discovered 6 of 423 characteristics were significantly linked to periodontitis (**Figure 5C**, **Supplementary File 2** — Figure s5). Notably, four of these high-confidence genes were discovered to be involved with multiple phenotypes: *S100A9*, *S100A12* (neutrophils and periodontitis); *MCM6*, *P14KAP2* (neutrophils and pDC) (**Table 3**, **Figure 5D**). Most of these significant features survived both conditional analysis and permutation testing (381/658 for neutrophils, 3/5 for NKT, 5/5 for pDC, and 5/6 for periodontitis). The majority of them were shown to be colocalized with their respective phenotype (554/658 for neutrophils, 4/5 for NKT, 3/5 for pDC, 0/6 for periodontitis), implying that shared and pleiotropic SNPs influence both gene expression and phenotype (**Supplementary File 1** — Table s12-15).

Discussion

Summary of key findings

In the present research, we employed MR to explore the potential links between circulating immune cells and periodontitis. Our study revealed causal relationships between elevated levels of circulating neutrophils, Natural Killer T cells, and plasmacytoid Dendritic Cells with a higher risk of periodontitis, despite the lack of robustness across sensitivity analyses. TWAS and colocalization analysis demonstrated possible cross-trait causal genes to be engaged in their interaction.

Circulating neutrophils play a significant part in periodontitis and inflammatory comorbidities

Notably, our findings suggested that circulating neutrophils may play a leading causal role in the likelihood of periodontitis, and it remained robust after correcting for potential confounding factors and outliers. Neutrophils are acknowledged as major actors in periodontitis since they

Trait	Ranking by MIP	MIP	MACE	Ranking by PP	PP	MSCE
Neutrophil	1	0.895	0.097	1	0.771	0.108
Natural Killer T cell	2	0.135	-0.003	2	0.056	-0.017
Plasmacytoid DC	3	0.102	0	3	0.045	0.007

Abbreviations: DC, dendritic cell; MACE, average causal effect of risk factor model; MIP, marginal inclusion probability; MR-BMA, Mendelian randomization-Bayesian model averaging; MSCE, model-specific causal estimates; PP, poster probability.

Table 2.

Ranking of risk factors and models for periodontitis in MR-BMA analysis.

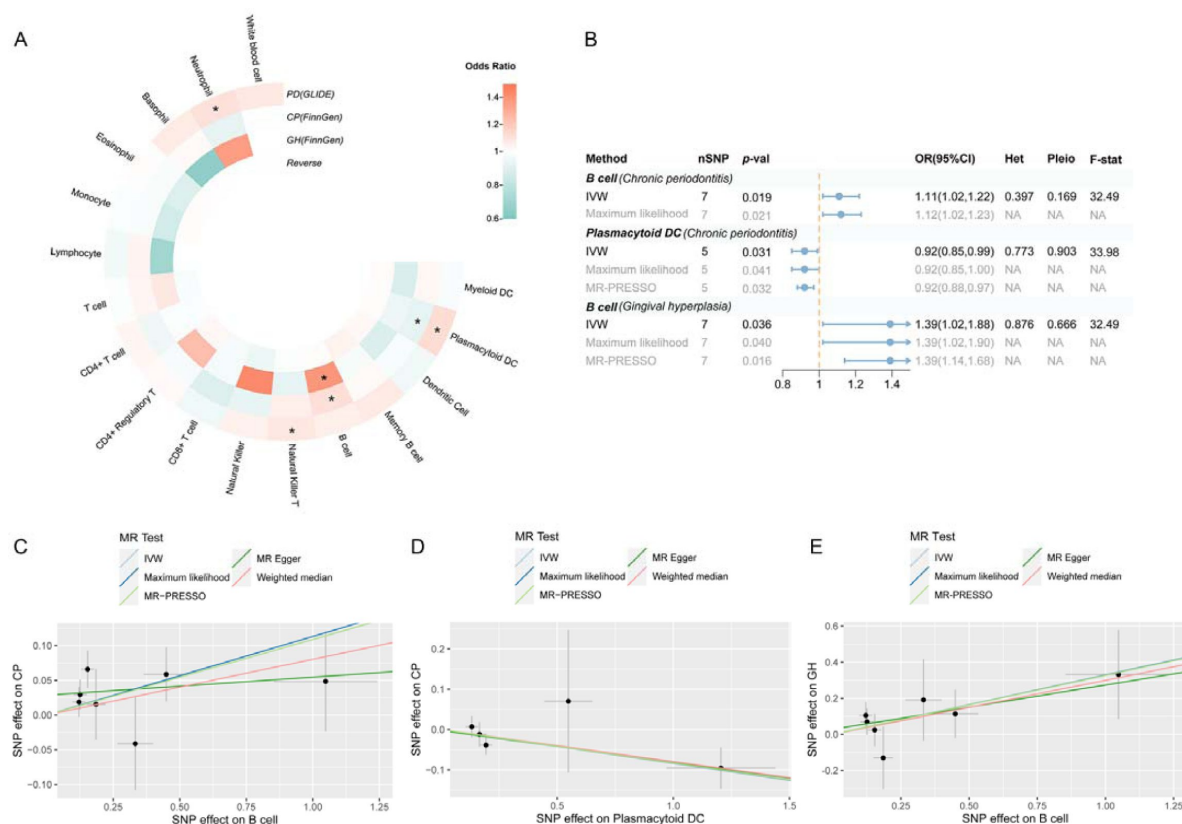


Figure 4.

Results of the subgroup and reverse MR. **(A)** A circular heatmap illustrates the results of the subgroup analysis and reverse MR. Lines in the heatmap represent periodontitis (GLIDE), chronic periodontitis (FinnGen), gingival hyperplasia (FinnGen), and reverse MR analysis, progressing from outside to inside. The color scale of the heatmap is determined by the odds ratio (OR). * $P < 0.05$. **(B)** A forest plot of the MR analyses for significant results in [Figure 4A](#) ($P < 0.05$). The effects are quantified using OR with 95% CI. **(C-E)** The effect estimates for each individual variant in B cell (CP) **(C)**, plasmacytoid DC (CP) **(D)**, and B cell (GH) **(E)** is provided by plotting SNP-outcome associations against SNP-exposure associations. The regression slope fitted by different MR methods is represented by lines with different colors.

Abbreviations: CI, confidence interval; DC, dendritic cell; F-stat, F-statistic; GLIDE, Gene-Lifestyle Interactions in Dental Endpoints collaboration consortium; IVW, inverse variance weighted; Het, heterogeneity; MR-PRESSO, Mendelian Randomization Pleiotropy RESidual Sum and Outlier; OR, odds ratio; PD, periodontitis; CP, chronic periodontitis; GH, gingival hyperplasia; Pleio, pleiotropy; SNP, single nucleotide polymorphism.

Gene name	Position	Phenotype	TWAS Z-score	TWAS <i>p</i> val	Perm <i>p</i> val	Model <i>p</i> val	PP.H3	PP.H4
<i>S100A9</i>	1q21.3	Neutrophil	7.225	5E-13	0	1.7E-17	0.035	0.962
		Periodontitis	-3.564	3.7E-4	1.3E-3	1.7E-17	0.008	0.259
<i>S100A12</i>	1q21.3	Neutrophil	7.31	2.7E-13	1.7E-4	4.1E-43	0.027	0.97
		Periodontitis	-3.513	4.4E-4	8.1E-4	4.1E-43	0.008	0.244
<i>MCM6</i>	2q21.3	Neutrophil	-5.549	2.9E-8	0	3.5E-13	0	0.999
		Plasmacytoid DC	7.172	7.4E-13	2.7E-3	3.5E-13	1	0
<i>PI4KAP2</i>	22q11.21	Neutrophil	5.345	9E-8	3.3E-3	3.7E-39	0.003	0.016
		Plasmacytoid DC	5.322	1E-7	3.2E-3	3.7E-39	0.009	0.045

Abbreviations: DC, dendritic cell; Perm, permutation test; PP, poster probability; TWAS, transcriptome-wide association study.

Table 3.

TWAS and colocalization analysis identified genes involved in multiple phenotypes.

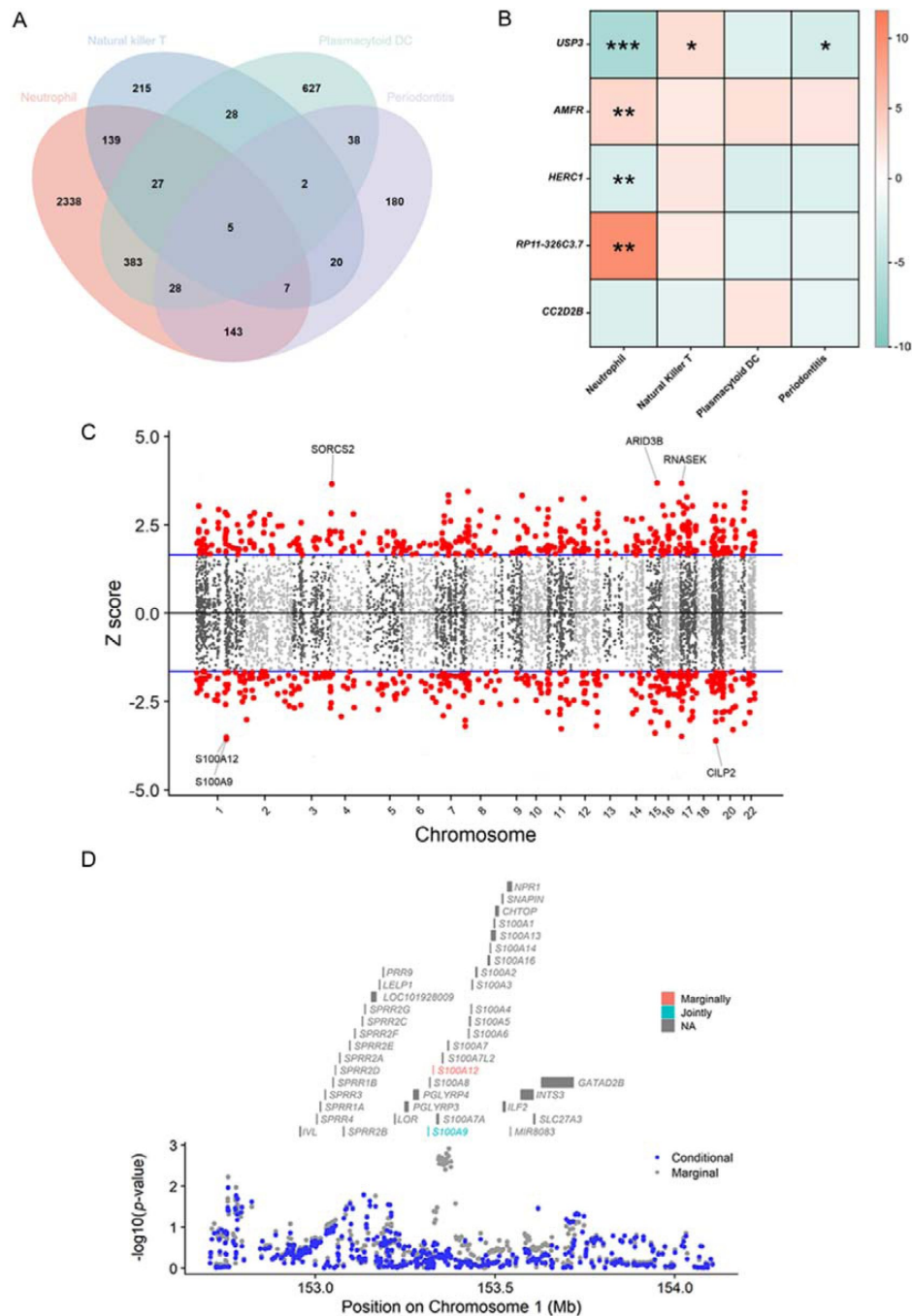


Figure 5.

Results of the TWAS and colocalization analysis. **(A)** A Venn diagram illustrates the intersecting genes shared by multiple traits ($P < 0.05$). **(B)** A heatmap representing the TWAS and colocalization analysis for five genes interacting among neutrophil, Natural Killer T cell, plasmacytoid DC, and periodontitis. The TWAS Z-score is used as the color scale for the heatmap. * PP.H3 + PP.H4 > 0.5; ** PP.H3 + PP.H4 > 0.8; *** PP.H4 > 0.8. **(C)** Manhattan plot of gene-trait associations for periodontitis. The X-axis represents genomic positions. Blue lines indicate a Z-score of 1.96. Red circles represent significant gene-trait associations ($P < 0.05$). Six genes satisfy a multiple corrected threshold of $P < 5 \times 10^{-4}$. **(D)** Regional Manhattan plot of conditional analysis for *S100A9*, *S100A12* in periodontitis. Grey bars indicate the location of genes on chromosome 1. Genes colored in orange and green on the graph indicate the marginally and jointly significant genes that best explain the GWAS signals. Gray and blue dots respectively indicate GWAS p -values before and after conditioning on the jointly significant gene. **Abbreviations:** DC, dendritic cell; GWAS, genome-wide association study; PP, posterior probability; TWAS, transcriptome-wide association study.

serve as the front line of host immune defense ([Ley et al., 2018](#)). Numerous pieces of clinical evidence have uncovered that neutrophils account for a significant portion of inflammatory tissue damage and that the severity of periodontitis is positively correlated with the overproduction, dysregulation, or hyperactivity of neutrophils ([Chapple et al., 2023](#); [Fine et al., 2021](#)). A case-control study indicated that periodontitis patients suffered from a higher level of apoptotic circulatory neutrophils than healthy people ([Nicu et al., 2018](#)). An increased neutrophil count could suggest the inflammatory burden of gingivitis and dental plaque in the oral cavity ([Sreenivasan and Prasad, 2022](#)). Another study discovered that neutrophil depletion ameliorated experimental periodontitis while unrestrained recruitment aggravated it ([Dutzan et al., 2018](#)).

Furthermore, neutrophils may play a significant role in the inflammatory comorbidities of periodontitis ([Hajishengallis and Chavakis, 2021](#)). Patients with severe periodontitis suffer from low-grade systemic inflammation, as evidenced by increased levels of circulating neutrophils and pro-inflammatory mediators compared to healthy controls ([D'Aiuto et al., 2013](#); [Schenkein et al., 2020](#)). Pre-clinical research also demonstrated that ligature-induced periodontitis (LIP) was accompanied by increased circulatory neutrophil counts, producing in endothelial dysfunction and vascular inflammation ([Brito et al., 2013](#)).

A recently developed concept known as “trained immunity” has opened up new avenues by which neutrophils promote periodontitis and comorbidities ([Li et al., 2022](#)). According to the theory, innate immune cell progenitors are able to recall the pathogens they encounter ([Netea et al., 2016](#)).

Periodontitis, as an example, stimulates myelopoiesis in bone marrow, leading to a rise of trained neutrophils in blood circulation and periodontal tissues ([Li et al., 2023](#)). These neutrophils cause an increase in the production of neutrophil extracellular traps (NETs) as well as a decrease in their degradation ([White et al., 2016](#)), which exacerbates epithelial barrier collapse and promotes bacteraemia ([Burmeister et al., 2022](#)).

Several lymphocyte subsets casually associated with the risk of periodontitis

NKT cells, a distinct fraction of T lymphocytes, are linked to the pathophysiology of a variety of inflammatory, osteolytic, and autoimmune diseases ([Godfrey et al., 2000](#)). Similar to our findings, previous research revealed a greater number of NKT recruited in periodontitis tissues ([Muthukuru, 2012](#); [Yamazaki et al., 2001](#)). Several studies have demonstrated the tissue-specific function of NKT and highlighted its pathogenic role in periodontitis ([Aoki-Nonaka et al., 2014](#); [Melgar-Rodríguez et al., 2021](#)), which may be attributed to the proinflammatory and immunoregulatory activities mediated by NKT, spanning from cytokine production to immune cell interactions ([Seidel et al., 2020](#)).

In addition, our study identified a convoluted causal relationship between pDC and periodontitis. Dendritic Cells, as specialized antigen-presenting cells, play a crucial role in the modulation of the host immune response and may be related to bone loss during periodontitis ([El-Awady et al., 2022](#); [Ginesin et al., 2023](#)). In response to viral encounters and infection, pDC represents a unique subgroup of DC that releases type I interferon (IFN) ([Jego et al., 2003](#)). However, pDC is only discovered in a tiny percentage of healthy oral tissues, and there remains a dearth of relevant clinical research ([Meghil and Cutler, 2020](#); [Wilensky et al., 2014](#)). The involvement of pDC in periodontitis deserves further investigation.

Systemic immunomodulation management for immune cells serves as a target for periodontal care

Periodontitis is a damaging inflammatory disease induced and exacerbated by the plaque biofilm and host immune response ([Moutsopoulos and Konkel, 2018](#)). The systemic immune response comprises both innate and adaptive immunity, with numerous cytokines, immune cells, and inflammatory pathways interacting in a complex crosstalk during periodontitis, hinting that immunomodulation management may be an essential target for periodontal care ([Dutzan et al., 2016](#); [Hajishengallis, 2014](#)).

Reactive periodontal therapies, which focus on plaque management, pocket depth reduction, and gingival bleeding eradication, do not always produce the intended results and fall short of a genuinely comprehensive approach to dental care ([Kornman et al., 2017](#)). Recently, a promising term “P4 periodontics” (Predictive, Preventive, Personalized, and Participatory) has been introduced as a multilayer healthcare paradigm for the management of periodontitis, emphasizing the personalized responsiveness of treatment to disease ([Bartold and Ivanovski, 2022](#)). Modulation of systemic host immune responses is particularly appropriate for predicting the progression and severity of periodontitis in persons whose periodontal condition is only slightly correlated with dental plaque ([Divaris et al., 2020](#)). MR contributes a novel approach to the investigation of systemic immunological alternations in periodontitis. A recent MR study evaluated the causal associations between circulating cytokines and the risk of periodontitis ([Huang et al., 2023](#)).

Our present study highlighted five genes (*USP3*, *AMFR*, *HERC1*, *CC2D2B*, and *RP11-326C3.7*) that may play a pivotal role in the communication between circulating neutrophils, pDC, NKT, and periodontitis, as well as two high confidence genes (*S100A9*, *S100A12*) situated within 1q21.3 as prospective gene targets for regulating circulating neutrophils during periodontitis. Our findings could pave the way for a novel preventive and therapeutic approach to modifying the systemic immunological equilibrium in periodontitis patients by modulating circulating immune cells. These findings may enable the prediction of individuals at risk of periodontitis through screening certain immune imbalances, which might then be employed to specifically prevent periodontitis and related inflammatory comorbidities, particularly in patients with systemic susceptibility factors.

Strengths and limitations in the present study

The present study exhibited several strengths. First, under the premise of three key assumptions, MR is a powerful tool for explaining the relationship between complicated features (such as circulating immune cells) by successfully mitigating the effect of probable confounders and allowing for reasonable causal order. Second, a rigorous quality control process was conducted in accordance with the STROBE-MR checklist in multiple domains, including IVs selection, heterogeneity investigations, and removal of pleiotropic loci. Third, we adopted a series of sensitivity tests and MVMR to rule out the impact of outlier, influential, or pleiotropic SNPs. Fourth, a novel method based on nonlinear Bayesian averaging was applied to explore the causal drivers of disease risk from a set of high-throughput risk factors. Finally, TWAS was used in conjunction with MR to identify achievable regulatory gene targets for periodontal care.

However, some limitations should be addressed when interpreting the results. To begin with, a scarcity of GWAS databases hampered more comprehensive and precise analyses. As a result, we were unable to evaluate the impact of immune cells on distinct subsets of periodontal illnesses (such as chronic gingivitis and periodontal abscess) or ethnic groups (such as East Asian and African). Second, the primary results from the IVW method were not stable across all alternative analyses, nor were they replicated within subgroups, implying that the findings had limited evidentiary power. Third, the two exposure datasets were incompatible in terms of sample size. Despite the fact that we explored several selection thresholds of IVs to reduce their influence on

results, the varied number of SNPs fulfilling the criteria may result in some bias. Fourth, none of the results satisfied the Bonferroni multiple correction ($P < 0.05/17 = 0.003$), which may have inflated the rate of type I errors. Fifth, in spite of our best attempts to minimize potential confounding factors, interferences from unobserved pleiotropies could not be completely ruled out. Sixth, in addition to quantities, function abnormalities (such as dysregulation or hyperactivity) of circulating immune cells may also be related to the susceptibility and severity of periodontitis, however our research failed to address this issue. Seventh, while the majority of immune cells in gingival crevicular fluid are derived from blood, the amount of circulating immune cells is influenced by more intricate factors, which may challenge the current causal inference. Finally, since MR evaluates causal inference from the standpoint of genetic variations, it may not always correspond exactly to fact.

Conclusions

In conclusion, the present study provides suggestive evidence of the casual associations of genetically predicted circulating neutrophils, Natural Killer T cells, and plasmacytoid Dendritic Cells on the risk of periodontitis, which shed light on the involvement of systemic immunological alterations in periodontitis etiology. Our findings may provide an innovative and evidence-based framework for the prospect of systemic immunomodulation management in periodontal care, which can be valuable for early diagnostics, risk assessment, targeted prevention, and personalized management of periodontitis, especially for patients with systemic susceptibility factors. However, the effect estimation discovered in our study was marginal, prompting caution when transferring to clinical practice. More studies are required to comprehend the biological plausibility and clinical applicability of our findings.

Data Availability

The data generated or analysed during this study are available in this published article and its supplementary information files. The data used in the study could be requested through the corresponding author with reasonable request.

<https://data.bris.ac.uk/data/dataset/>

<https://finngen.gitbook.io/documentation/v/r9/data-download>

<https://gtexportal.org/home>

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Competing interests

The authors declare no competing interests.

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Ethics

Ethics approval and consent to participate are not applicable.

Additional files

Supplementary files

Supplementary File 1. Tables s1-15.

Supplementary File 2. Figures s1-5.

Data availability

Data from GLIDE can be obtained via application (<https://data.bris.ac.uk/data/dataset/>), FinnGen can be obtained via application (<https://finngen.gitbook.io/documentation/v/r9/data-download>), and GTEx can be obtained via application (<https://gtexportal.org/home>). The data generated or analysed during this study are available in this published article and its supplementary information files. The data used in the study could be requested through the corresponding author with reasonable request.

Code availability

The software and code used in the study could be obtained through the corresponding author.

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Editors

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Reviewer #1 (Public Review):

Ye et al. used Mendelian randomization method to evaluate the causative association between circulating immune cells and periodontitis and finally screened out three risk immune cells related to periodontitis. Overall, this is an important and novel piece of work that has the potential to contribute to our understanding of the causal relationship between circulating immune cells related to periodontitis. However, there are still some concerns that need to be addressed.

1. The authors used $1e-9$ as the threshold to select effective instrumental variables (IVs), which should give the corresponding references. Meanwhile, the authors should test and discuss the potential impact of inconsistent thresholds for exposure ($1e-9$, $5e-6$ were selected by the author respectively) and outcome IVs ($5e-8$) on the robustness of the results.
2. What is the reference for selecting Smoking, Fasting plasma glucose, and BMI as covariates? They do not seem to be directly related to immune cells as confounding factors.
3. It is not entirely clear about the correction of P-value for the total number of independent statistical tests.
4. The author used whole blood data to apply FUSION algorithm. Although whole blood is a representative site, the authors should add FUSION testing of periodontally relevant tissues, such as oral mucosa.
5. The authors chose gingival hyperplasia as a secondary validation phenotype of periodontitis in this study. However, gingival recession, as another important phenotype associated with periodontitis, should also be tested and discussed.
6. This study used GLIDE data as a replicated validation, but the results were inconsistent with FinnGen's dataset.

Reviewer #2 (Public Review):

This manuscript presents a well-designed study that combines multiple Mendelian randomization analyses to investigate the causal relationship between circulating immune cells and periodontitis. The main conclusions of the manuscript are appropriately supported by the statistics, and the methodologies used are comprehensive and rigorous.

These findings have significant implications for periodontal care and highlight the potential for systemic immunomodulation management on periodontitis, which is of interest to readers in the fields of periodontology, immunology, and epidemiology.